Figure 12.5 A typical clustering produced by k-means. Dataset's points are plotted as circles, centroids are shown as thick-edge Xs.

First, let's start with a positive note about k-means: figure 12.5 shows how it manages to correctly identify clusters with different density, for instance the red and orange ones in the bottom-left corner have a larger average distance between points (and hence a lower density) than the cyan cluster in the top-right corner. This might seem something to take for granted, but not all clustering algorithms will deal so well with heterogeneous distributions: in the next section, for example, we'll see why this is a problem for DBSCAN.